

# EMERSON EIMS

## BUILDING SUITE PRO

### New Project

Location: Nairobi, Kenya  
Total Area: 100 m² | Units: 1 | Storeys: 1  
Architectural Style: Modern  
Date: 26 April 2026

| PROJECT INFORMATION |                                      |
|---------------------|--------------------------------------|
| Project ID          | 3b5aff5c-5b68-4085-8e04-24b2a46e847f |
| Location            | Nairobi, Kenya                       |
| Total Area          | 100 m²                               |
| Units / Apartments  | 1                                    |
| Storeys             | 1                                    |
| Bedrooms per Unit   | 1                                    |
| Foundation Type     | Pad                                  |
| Style               | Modern                               |
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## VALIDATION ENGINE — AUDIT TRAIL

■ 3 BLOCKER(S) — DO NOT SUBMIT

- Location is mandatory — county submission cannot proceed without a recognised locality.
- Total floor area is missing or non-positive.
- Kenya statutory IDs missing: plot\_no, architect, engineer, qs. Required for BORAQS / EBK / NCA / county submission.

7 auto-correction(s) applied with code citation:

| Field      | Original      | Auto-corrected to | Reason / Code citation                                   |
|------------|---------------|-------------------|----------------------------------------------------------|
| Location   | (empty)       | Nairobi, Kenya    | Defaulted to capital — REPLACE before submission         |
| Currency   | USD           | KES               | CBK Cap.491 — statutory billing currency in Kenya is KES |
| Area (m²)  | 0             | 100               | Placeholder — REPLACE before submission                  |
| Stories    | 0             | 1                 | Minimum 1 storey enforced                                |
| Units      | 0             | 1                 | Minimum 1 unit enforced                                  |
| Bedrooms   | 0             | 1                 | Minimum 1 bedroom enforced                               |
| Foundation | (unspecified) | pad               | Single-storey ≤ 120 m² → pad (BS 8004 §7.3)              |

**Compliance Status: BLOCKED — DO NOT SUBMIT**

# 1. DESIGN BRIEF & DESIGN CRITERIA

| Item                         | Value                      |
|------------------------------|----------------------------|
| Project name                 | New Project                |
| Location                     | Nairobi, Kenya             |
| Total floor area             | 100.0 m²                   |
| No. of storeys / floor count | 1 (single source of truth) |
| Dwelling units               | 1                          |
| Bedrooms / unit              | 1                          |
| Architectural style          | Modern                     |
| Foundation                   | PAD                        |
| Currency                     | KES                        |
| Plot / LR No.                | — REQUIRED —               |
| Coordinates (lat,lng)        | —                          |
| Title Deed                   | —                          |
| Architect (BORAQS)           | — REQUIRED —               |
| Structural Engineer (EBK)    | — REQUIRED —               |
| Quantity Surveyor (BORAQS)   | — REQUIRED —               |
| Client                       | —                          |

## Design loads (BS EN 1991-1-1 Cat A — residential):

| Action                                                | Value (kN/m²) | Reference           |
|-------------------------------------------------------|---------------|---------------------|
| Permanent action $g_k$ (slab + finishes + partitions) | 7.00          | EC1 Table 6.1       |
| Variable action $q_k$                                 | 2.00          | EC1 Table 6.2 Cat A |
| Design ULS $w = 1.35 g_k + 1.5 q_k$                   | 12.45         | EC0 §6.4.3.2        |
| Design SLS $w = g_k + q_k$                            | 9.00          | EC0 §6.5.3          |

## 2. STRUCTURAL DESIGN CALCULATIONS

Calculations to BS EN 1992-1-1 (EC2) using  $f_{ck} = 30 \text{ MPa}$ ,  $f_{yk} = 500 \text{ MPa}$ , cover per exposure class XC1 internal / XC3 external. All members hand-verified against simplified Eurocode 2 design rules; final detailing to be reviewed by the Structural Engineer of Record (EBK-registered) before issue for construction.

### 2.1 Slab design (one/two-way RC slab)

| Parameter                 | Value                     |
|---------------------------|---------------------------|
| Short span $L_x$          | 9.13 m                    |
| Long span $L_y$           | 10.95 m                   |
| Behaviour                 | Two-way                   |
| Slab thickness $h$        | 350 mm (span/depth = 26)  |
| Effective depth $d$       | 319.0 mm                  |
| Cover                     | 25 mm (XC1)               |
| Design moment $M$         | 64.33 kN·m / m            |
| As required               | 488.0 mm <sup>2</sup> / m |
| As minimum (EC2 §9.3.1.1) | 455.0 mm <sup>2</sup> / m |
| As provided               | 488.0 mm <sup>2</sup> / m |
| Reinforcement             | T12 @ 225 c/c BW          |

### 2.2 Beam design (simply-supported RC beam)

| Parameter                 | Value                         |
|---------------------------|-------------------------------|
| Clear span $L$            | 5.48 m                        |
| Beam section $b \times h$ | 225 × 450 mm                  |
| Effective depth $d$       | 404.0 mm                      |
| Cover                     | 30 mm                         |
| Service load $w$          | 59.36 kN/m                    |
| Design moment $M$         | 222.6 kN·m                    |
| Design shear $V$          | 162.6 kN                      |
| As required (tension)     | 1333.0 mm <sup>2</sup>        |
| Main steel                | 7 no. T16 bot · 4 no. T16 top |
| Shear links               | T8 @ 300 c/c                  |

### 2.3 Column design (axially loaded RC column)

| Parameter            | Value                                |
|----------------------|--------------------------------------|
| Axial load $N$ (ULS) | 316.0 kN                             |
| Section              | 225 × 225 mm                         |
| Concrete grade       | C25/30 ( $f_{ck} = 30 \text{ MPa}$ ) |
| Steel grade          | $f_{yk} = 500 \text{ MPa}$           |
| Steel ratio $\rho$   | 0.020 (2.0 %)                        |
| As required          | 1012.0 mm <sup>2</sup>               |
| Main steel           | 6 no. T16                            |
| Links                | T8 @ 192 c/c                         |

### 2.4 Foundation design (per BS 8004 / EN 1997-1)

| Parameter                        | Value                      |
|----------------------------------|----------------------------|
| Foundation type                  | PAD                        |
| Plan size                        | 1.5 × 1.5                  |
| Thickness                        | 750 mm                     |
| Concrete grade                   | C25/30                     |
| Cover                            | 75 mm (XC2 ground contact) |
| Safe bearing pressure (presumed) | 150 kPa                    |
| Column load to foundation        | 316.0 kN                   |
| Reinforcement                    | T12 @ 150 c/c BW bottom    |

■ The bearing pressure shown is a BS 8004 Table 1 \*presumed\* value pending a site investigation. A signed Geotechnical Investigation Report (boreholes, SPT, allowable bearing capacity) must be obtained from a geotechnical engineer before construction. EIMS records this gap honestly rather than fabricating SPT values.

### 3. BAR BENDING SCHEDULE (BS 8666)

Schedule of all reinforcement bars with mark, diameter, shape code, length per bar, number off, total length and weight. Steel grade 500B. Bend radii and bar shapes comply with BS 8666:2020 Table 2. Quantities to be verified by the Engineer before cutting and bending.

| Bar Mark | Dia. | Shape Code | Length per bar (m) | No. off | Total length (m) | Weight (kg) |
|----------|------|------------|--------------------|---------|------------------|-------------|
| S1       | T12  | 00         | 10.30              | 80      | 824.00           | 731.7       |
| B1-bot   | T16  | 00         | 5.98               | 48      | 286.91           | 453.0       |
| B1-top   | T16  | 37         | 5.98               | 16      | 95.64            | 151.0       |
| B1-link  | T8   | 51         | 0.85               | 144     | 122.40           | 48.3        |
| C1       | T16  | 00         | 3.50               | 24      | 84.00            | 132.6       |
| C1-link  | T8   | 51         | 0.60               | 72      | 43.20            | 17.1        |
| F1       | T12  | 00         | 3.00               | 48      | 144.00           | 127.9       |
|          |      |            |                    |         | TOTAL            | 1661.6      |

**Total reinforcement steel: 1,662 kg  $\approx$  1.66 tonnes**

## 4. MEP DESIGN — ELECTRICAL, PLUMBING & DRAINAGE

Electrical design per IEC 60364 / KS IEC 60364. Plumbing & sanitary drainage per BS EN 12056-2 and Kenya NCC Building Code SEHS. Septic tank to NEMA EIA Regulations 2003.

### 4.1 Electrical load schedule

| Load description                          | Value             |
|-------------------------------------------|-------------------|
| Lighting (LED) @ 5 W/m <sup>2</sup>       | 500 W             |
| Small power sockets @ 25 W/m <sup>2</sup> | 2500 W            |
| Cooker / hob (per unit)                   | 6000 W            |
| Water heaters                             | 3000 W            |
| Air conditioning (split units)            | 1500 W            |
| TOTAL CONNECTED LOAD                      | 13,500 W          |
| Diversity factor                          | 0.65              |
| DESIGN MAXIMUM DEMAND                     | 8,775 W (9.2 kVA) |
| Main breaker (single phase 230 V)         | 45 A              |

### 4.2 Final-circuit schedule

| Circuit             | Load   | Cable                   | Protection       | RCD       |
|---------------------|--------|-------------------------|------------------|-----------|
| L1-Ln (Lighting)    | 500 W  | 1.5 mm <sup>2</sup> T+E | 10 A B-curve MCB | —         |
| P1-Pn (Sockets)     | 2500 W | 2.5 mm <sup>2</sup> T+E | 20 A B-curve MCB | 30 mA RCD |
| C (Cooker)          | 6000 W | 6 mm <sup>2</sup> T+E   | 32 A B-curve MCB | —         |
| W (1× water heater) | 3000 W | 4 mm <sup>2</sup> T+E   | 25 A B-curve MCB | 30 mA RCD |
| A1-A1 (AC splits)   | 1500 W | 2.5 mm <sup>2</sup> T+E | 16 A C-curve MCB | —         |

### 4.3 Single-line diagram (text representation)

KPLC supply ■ kWh meter ■ Main isolator 45 A ■ Main DB (Type-B SPD)  
■ Lighting circuits L1...Ln (1.5 mm<sup>2</sup> · 10 A)  
■ Socket circuits P1...Pn (2.5 mm<sup>2</sup> · 20 A · 30 mA RCD)  
■ Cooker C (6 mm<sup>2</sup> · 32 A)  
■ Water heaters W (4 mm<sup>2</sup> · 25 A · 30 mA RCD)  
■ AC splits A1...An (2.5 mm<sup>2</sup> · 16 A)  
■ Earth busbar ■ copper earth-rod ≤ 1 Ω

### 4.4 Plumbing fixture schedule

| Fixture           | Qty |
|-------------------|-----|
| Water Closet (WC) | 2   |
| Wash-Hand Basin   | 2   |
| Shower / Bath     | 2   |
| Kitchen Sink      | 1   |
| Laundry Sink      | 1   |
| External Bib Tap  | 2   |

### 4.5 Drainage gradient table (BS EN 12056-2)

| Pipe                  | Slope range  | Min fall              |
|-----------------------|--------------|-----------------------|
| 100 mm soil pipe (WC) | 1:40 to 1:80 | ≥ 1.00 m fall per 1 m |
| 75 mm waste           | 1:40         | 0.025 m / m           |

|                  |      |             |
|------------------|------|-------------|
| 50 mm waste      | 1:40 | 0.025 m / m |
| 32 mm WHB / sink | 1:20 | 0.050 m / m |

#### 4.6 Septic tank & soak-pit sizing

| Parameter                                           | Value                              |
|-----------------------------------------------------|------------------------------------|
| Design occupancy                                    | 2 persons (BS 8233 — bedrooms × 2) |
| Daily wastewater flow (180 L/p/d)                   | 360 L/day                          |
| Septic tank volume (3-day retention)                | 1.08 m <sup>3</sup>                |
| Suggested tank dimensions                           | 2.5 × 1.2 × 0.36 m (L×W×D)         |
| Soak-pit area (porous soil, 30 L/m <sup>2</sup> /d) | 12.0 m <sup>2</sup>                |
| Soak-pit diameter (circular)                        | 3.91 m                             |

■ Final septic-system geometry must be confirmed by an on-site percolation test. A NEMA Environmental Impact Assessment (EIA) project report is required for systems > 50 persons or sites near surface watercourses.

## 5. BILL OF QUANTITIES (KQS-2025)

Contractor-grade BOQ in KES, structured by trade per the Kenya Quantity Surveyors Standard Method of Measurement. Rates are KQS-2025 indicative urban rates; tenderers shall price all items. All quantities to be verified on site.

### A — PRELIMINARIES

| Description                          | Unit | Qty  | Rate           | Amount             |
|--------------------------------------|------|------|----------------|--------------------|
| Site mobilisation, hoarding, signage | item | 1.00 | KSh 180,000    | KSh 180,000        |
| Insurance & NCA Levy                 | item | 1.00 | KSh 220,000    | KSh 220,000        |
| Site office + storage                | item | 1.00 | KSh 95,000     | KSh 95,000         |
|                                      |      |      | Trade subtotal | <b>KSh 495,000</b> |

### B — SUBSTRUCTURE

| Description                         | Unit | Qty    | Rate           | Amount             |
|-------------------------------------|------|--------|----------------|--------------------|
| Site clearance & strip topsoil      | m²   | 130.00 | KSh 120        | KSh 15,600         |
| Bulk excavation foundation trenches | m³   | 30.00  | KSh 650        | KSh 19,500         |
| Hardcore filling 200 mm             | m²   | 100.00 | KSh 850        | KSh 85,000         |
| Lean concrete blinding 50 mm        | m³   | 5.00   | KSh 14,000     | KSh 70,000         |
| RC foundation C25/30                | m³   | 30.00  | KSh 26,000     | KSh 780,000        |
|                                     |      |        | Trade subtotal | <b>KSh 970,100</b> |

### C — SUPERSTRUCTURE

| Description                      | Unit | Qty      | Rate           | Amount               |
|----------------------------------|------|----------|----------------|----------------------|
| RC columns C25/30                | m³   | 0.71     | KSh 32,500     | KSh 23,034           |
| RC beams C25/30                  | m³   | 4.44     | KSh 30,500     | KSh 135,315          |
| RC suspended slabs C25/30        | m³   | 35.00    | KSh 28,500     | KSh 997,500          |
| Reinforcement bars (BBS)         | kg   | 1,661.60 | KSh 165        | KSh 274,164          |
| Formwork to soffits & sides      | m²   | 240.87   | KSh 1,450      | KSh 349,264          |
| 200 mm machine-cut block walling | m²   | 120.00   | KSh 1,800      | KSh 216,000          |
| Plaster both sides               | m²   | 240.00   | KSh 1,100      | KSh 264,000          |
|                                  |      |          | Trade subtotal | <b>KSh 2,259,277</b> |

### D — ROOFING

| Description                                | Unit | Qty    | Rate           | Amount             |
|--------------------------------------------|------|--------|----------------|--------------------|
| Timber roof trusses & purlins              | m²   | 100.00 | KSh 2,200      | KSh 220,000        |
| 0.6 mm box-profile coloured iron sheet     | m²   | 105.00 | KSh 1,450      | KSh 152,250        |
| 9 mm gypsum ceiling on softwood brandering | m²   | 100.00 | KSh 1,850      | KSh 185,000        |
|                                            |      |        | Trade subtotal | <b>KSh 557,250</b> |

### E — FINISHES

| Description                               | Unit | Qty    | Rate      | Amount      |
|-------------------------------------------|------|--------|-----------|-------------|
| Floor ceramic / porcelain tiling          | m²   | 85.00  | KSh 2,400 | KSh 204,000 |
| Wall tiling (wet areas)                   | m²   | 18.00  | KSh 2,200 | KSh 39,600  |
| Internal paint (Crown / Plascon emulsion) | m²   | 192.00 | KSh 520   | KSh 99,840  |
| External paint (weather-shield)           | m²   | 96.00  | KSh 720   | KSh 69,120  |

|  |  |  |                |                    |
|--|--|--|----------------|--------------------|
|  |  |  | Trade subtotal | <b>KSh 412,560</b> |
|--|--|--|----------------|--------------------|

## F — JOINERY

| Description                        | Unit | Qty   | Rate           | Amount             |
|------------------------------------|------|-------|----------------|--------------------|
| Internal flush doors with hardware | each | 4.00  | KSh 18,000     | KSh 72,000         |
| External main doors hardwood       | each | 2.00  | KSh 42,000     | KSh 84,000         |
| Aluminium-framed sliding windows   | m²   | 15.00 | KSh 9,500      | KSh 142,500        |
|                                    |      |       | Trade subtotal | <b>KSh 298,500</b> |

## G — ELECTRICAL

| Description                                 | Unit  | Qty   | Rate           | Amount             |
|---------------------------------------------|-------|-------|----------------|--------------------|
| Mains supply, DB, MCB, RCDs (per IEC 60364) | item  | 1.00  | KSh 165,000    | KSh 165,000        |
| Lighting points & circuits (~500 W)         | point | 18.00 | KSh 2,200      | KSh 39,600         |
| Socket outlets (~2500 W)                    | point | 12.00 | KSh 2,400      | KSh 28,800         |
|                                             |       |       | Trade subtotal | <b>KSh 233,400</b> |

## H — PLUMBING

| Description                                 | Unit | Qty  | Rate           | Amount             |
|---------------------------------------------|------|------|----------------|--------------------|
| Cold + hot water reticulation, all fixtures | item | 1.00 | KSh 285,000    | KSh 285,000        |
| Septic tank + soak-pit (1.08 m³ + Ø3.91 m)  | item | 1.00 | KSh 195,000    | KSh 195,000        |
| Stormwater drainage & rain-water disposal   | item | 1.00 | KSh 145,000    | KSh 145,000        |
|                                             |      |      | Trade subtotal | <b>KSh 625,000</b> |

## Cost summary

| Item                                              | Amount          |
|---------------------------------------------------|-----------------|
| A — PRELIMINARIES                                 | KSh 495,000     |
| B — SUBSTRUCTURE                                  | KSh 970,100     |
| C — SUPERSTRUCTURE                                | KSh 2,259,277   |
| D — ROOFING                                       | KSh 557,250     |
| E — FINISHES                                      | KSh 412,560     |
| F — JOINERY                                       | KSh 298,500     |
| G — ELECTRICAL                                    | KSh 233,400     |
| H — PLUMBING                                      | KSh 625,000     |
| Subtotal of works                                 | KSh 5,851,087   |
| Contingency 7.5 %                                 | KSh 438,832     |
| Professional fees 8 % (Architect + Engineer + QS) | KSh 468,087     |
| Sub-total before VAT                              | KSh 6,758,006   |
| VAT 16 %                                          | KSh 1,081,281   |
| GRAND TOTAL                                       | KSh 7,839,287   |
| Cost per m² (KES)                                 | KSh 78,393 / m² |

## 6. STATUTORY COMPLIANCE PACK

All approvals required under Kenyan law before construction. The contractor and professional team shall ensure each item is closed prior to mobilisation.

| Approval                         | Authority / Statute                      | Documents required                                                                |
|----------------------------------|------------------------------------------|-----------------------------------------------------------------------------------|
| 1. County Planning Approval      | County Govt (Building Plans Section)     | Stamped architectural & structural drawings, location plan, ownership docs, NCA   |
| 2. NCA Project Registration      | National Construction Authority Act 2011 | NCA-1 form, contractor registration certificate, levy 0.5 % of contract sum.      |
| 3. NEMA Project Report (PR)      | EMCA 1999, EIA Regs 2003                 | Required for projects > 5 ha or near watercourses; PR submitted to NEMA county    |
| 4. Public Health Approval        | Public Health Act Cap. 242               | Septic system & water supply review by County Public Health Office.               |
| 5. EPRA / KPLC Energy Connection | EPRA Regulations                         | Electrical drawings & wayleave application; final inspection by EPRA-registered C |
| 6. Fire Safety Certificate       | County Fire Marshal                      | Means-of-escape, extinguisher schedule, hydrant layout (NFPA 241 / Kenya Fire     |
| 7. DOSHS Notification            | OSHA 2007 §41                            | Notification before commencement; site safety plan, induction registers, ISO 4500 |

### 6.1 Professional certification (signature pages)

| Role                       | Name & Reg.         | Signature & Stamp | Date        |
|----------------------------|---------------------|-------------------|-------------|
| Architect (BORAQS)         | — REQUIRED —        | _____             | Date: _____ |
| Structural Engineer (EBK)  | — REQUIRED —        | _____             | Date: _____ |
| Quantity Surveyor (BORAQS) | — REQUIRED —        | _____             | Date: _____ |
| Site Manager (NCA)         | — to be appointed — | _____             | Date: _____ |
| Client                     | —                   | _____             | Date: _____ |

I certify that the design has been carried out in accordance with the relevant Kenyan codes (NCA Act 2011, EMCA 1999, OSHA 2007) and applicable Eurocodes / British Standards, and that I take professional responsibility for the section of work above corresponding to my registration.

## 8. PROJECT SUMMARY

### Phase phase\_1: Site Analysis

|                     |                                                                                  |
|---------------------|----------------------------------------------------------------------------------|
| water_table_depth   | 3.5                                                                              |
| water_table_source  | regional_estimate                                                                |
| water_table_note    | Regional estimate from climate/soil model -- verify with on-site geotechnical su |
| soil_type           | clay                                                                             |
| vegetation_coverage | 0.65                                                                             |
| slope_angle         | 8.25                                                                             |
| suitability_pct     | 90                                                                               |
| nearby_structures   | survey_required                                                                  |
| land_use            | residential                                                                      |
| climate_zone        | tropical                                                                         |
| analysis_method     | regional_data_model                                                              |
| data_source         | EIMS Regional Database + GPS Coordinates                                         |

### Phase phase\_2: Geotechnical Design

|                        |        |
|------------------------|--------|
| bearing_capacity_kPa   | 150    |
| soil_classification    | CH     |
| recommended_foundation | STRIP  |
| settlement_mm          | 40     |
| slope_stability        | stable |
| water_table_depth      | 3.5    |
| excavation_depth       | 2.8    |

### Phase phase\_3: Foundation Design

|                        |                      |
|------------------------|----------------------|
| foundation_type        | PAD                  |
| foundation_depth_m     | 1.3                  |
| footing_width_m        | 2.2                  |
| total_building_load_kN | 3375.0               |
| bearing_capacity_kPa   | 150                  |
| safety_factor          | 3.0                  |
| concrete_volume_m3     | 87.75                |
| reinforcement_steel_kg | 7897.5               |
| soil_type              | clay                 |
| water_table_factor     | standard             |
| design_standard        | BS 8004 / Eurocode 7 |
| bim_ready              | True                 |

### Phase phase\_4: Floor Plans

|                |                                                                                 |
|----------------|---------------------------------------------------------------------------------|
| floor_count    | 2                                                                               |
| total_area_sqm | 450.0                                                                           |
| bathrooms      | 6                                                                               |
| rooms          | {'bedrooms': 9, 'bathrooms': 6, 'living_areas': 3, 'kitchen': 3, 'extra_rooms': |

|                          |                             |
|--------------------------|-----------------------------|
| style                    | modern                      |
| area_per_unit            | 150.0                       |
| building_code_compliance | True                        |
| area_efficiency_percent  | 77.5                        |
| cost_per_sqm_estimate    | calculated_in_costing_phase |

### ***Phase phase\_5: Electrical Design***

|                          |                   |
|--------------------------|-------------------|
| total_connected_load_kW  | 24.9              |
| max_demand_kW            | 12.45             |
| diversity_factor         | 0.5               |
| supply_type              | single-phase 230V |
| main_current_A           | 54.1              |
| lighting_points          | 30                |
| socket_outlets           | 30                |
| total_circuits           | 17                |
| lighting_circuits        | 3                 |
| socket_circuits          | 5                 |
| fixed_appliance_circuits | 9                 |
| distribution_boards      | 3                 |

### ***Phase phase\_6: Plumbing Design***

|                          |      |
|--------------------------|------|
| total_occupants          | 13   |
| daily_demand_liters      | 1950 |
| peak_hourly_liters       | 195  |
| total_fixture_units      | 72.0 |
| main_supply_pipe_mm      | 32   |
| cold_water_pipes         | 24   |
| hot_water_pipes          | 15   |
| waste_outlets            | 21   |
| soil_stacks              | 1    |
| hot_water_storage_liters | 520  |
| water_heater_kW          | 20.8 |
| storage_tank_liters      | 1950 |

### ***Phase phase\_7: BOQ***

|                      |                                                                                  |
|----------------------|----------------------------------------------------------------------------------|
| material_items       | 77                                                                               |
| total_material_cost  | 127953.59                                                                        |
| currency             | USD                                                                              |
| items_by_category    | {'mep_hvac': 5, 'electrical': 11, 'plumbing': 14, 'finishes': 18, 'structural':  |
| items                | [{'item_no': 1, 'description': 'Portland Cement (50kg)', 'unit': 'bag', 'quantit |
| calculation_basis    | {'total_area_m2': 450.0, 'stories': 2, 'units': 3, 'bedrooms_per_unit': 3, 'foot |
| waste_factor_applied | True                                                                             |
| method               | quantity_takeoff_from_dimensions                                                 |

### ***Phase phase\_8: Infrastructure***

|                             |                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------|
| solar_system                | {'panels_kw': 9.0, 'panels_count': 23, 'battery_kWh': 13.5, 'estimated_cost_usd': 1676.88}  |
| borehole_system             | {'depth_meters': 8, 'estimated_yield_lpm': 15, 'pump_kw': 3, 'estimated_cost_usd': 1676.88} |
| grid_option                 | {'connection_cost_usd': 375, 'monthly_tariff_estimate_usd': 67}                             |
| recommendation              | solar_grid_hybrid                                                                           |
| estimated_annual_saving_usd | 1676.88                                                                                     |

### Phase phase\_9: Landscape Design

|                         |               |
|-------------------------|---------------|
| lawn_area_m2            | 157.5         |
| hardscape_area_m2       | 45.0          |
| garden_area_m2          | 67.5          |
| plants_count            | 540           |
| irrigation_system       | True          |
| recommended_style       | tropical_lush |
| estimated_cost_usd      | 1552          |
| maintenance_monthly_usd | 31            |

### Phase phase\_10: Permits & Compliance

|                          |                                                                                      |
|--------------------------|--------------------------------------------------------------------------------------|
| jurisdiction             | Other                                                                                |
| building_code            | International Building Code                                                          |
| building_code_compliance | REQUIRES_VERIFICATION                                                                |
| permits_required         | 6                                                                                    |
| estimated_timeline_days  | 45                                                                                   |
| estimated_fees_usd       | 720                                                                                  |
| setback_compliance       | {'front_m': 5, 'side_m': 2, 'rear_m': 3}                                             |
| required_documents       | ['Architectural drawings (stamped)', 'Structural calculations (engineer-certified)'] |

### Phase phase\_11: 3D Visualization

|                |                        |
|----------------|------------------------|
| status         | ready                  |
| model_endpoint | /api/drawings/3d-model |

### Phase phase\_12: Live Costing

|                    |           |
|--------------------|-----------|
| material_cost      | 127953.59 |
| labor_cost         | 28789.56  |
| equipment_cost     | 10236.29  |
| overhead           | 18809.18  |
| contingency        | 22294.63  |
| subtotal           | 208083.25 |
| contractor_profit  | 41616.65  |
| total_project_cost | 249699.9  |
| cost_per_sqm       | 554.89    |
| labor_multiplier   | 0.5       |
| region             | Other     |
| currency           | USD       |

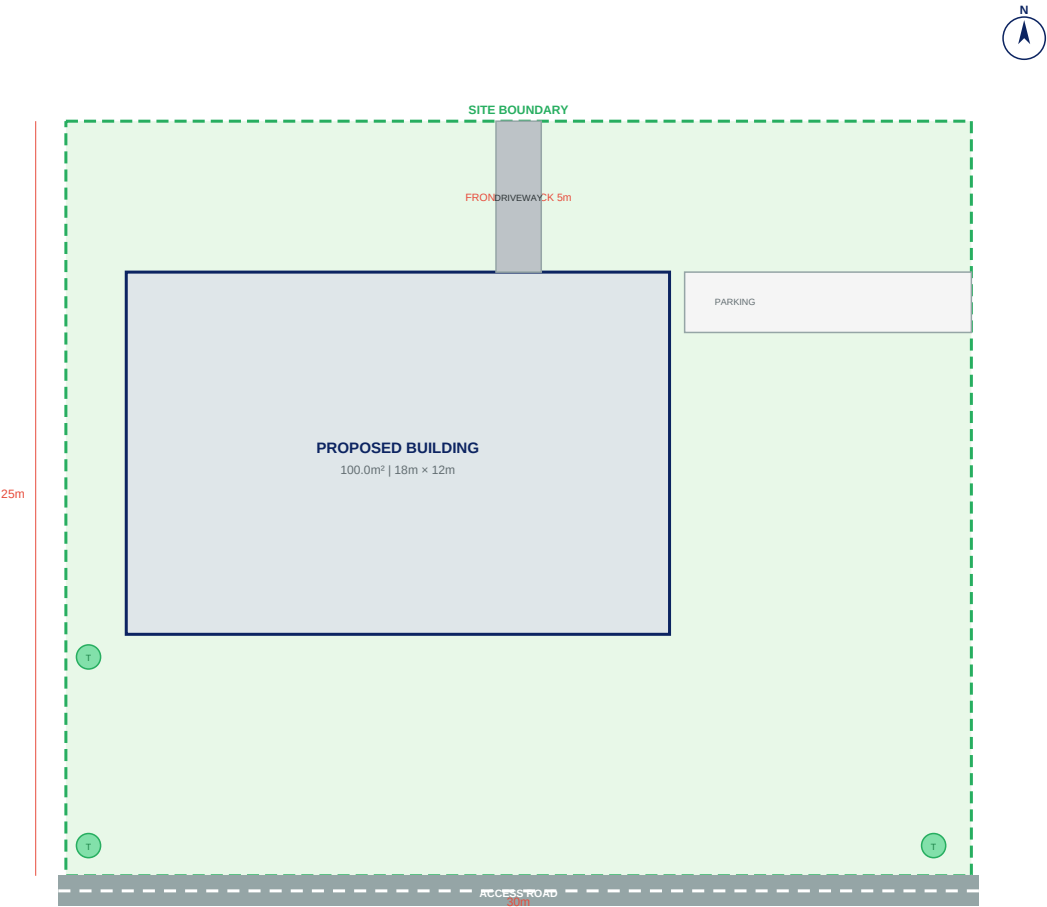
### Phase phase\_13: Integration Layer

|                |                                         |
|----------------|-----------------------------------------|
| status         | complete                                |
| export_formats | ['IFC2x3', 'FBX', 'DXF', 'PDF', 'XLSX'] |

# 9. SITE PLAN

Site layout showing building footprint, access roads, setbacks and landscaping. Area: 100 m²

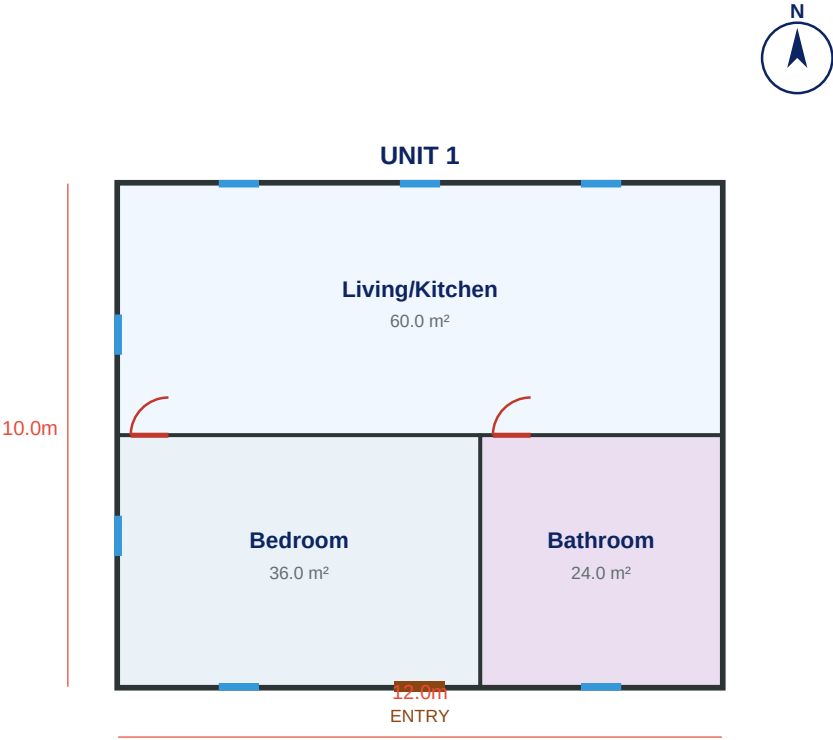
SITE PLAN - Nairobi, Kenya | Plot: 30m×25m



# 10. ARCHITECTURAL FLOOR PLAN

Ground floor layout — 1 unit(s), 1 bedrooms, showing room dimensions, doors, windows and circulation.

## FLOOR PLAN - 100.0m² | 1 Units | 1 Storey(s)



EMERSON EIMS BUILDING SUITE PRO

Area: 100.0m² | 1 Units × 1 Bed

Floor Plan | Scale 1:50 | 2026-04-26 | Drawing: FP-001

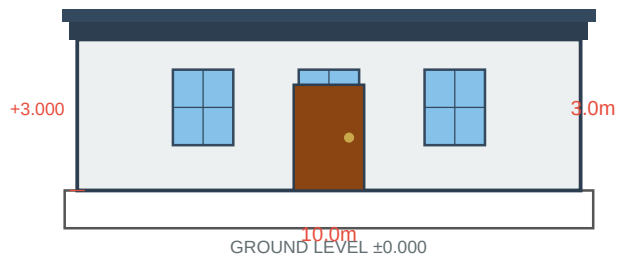
IFC Ready | BIM Level 2

# 11. BUILDING ELEVATIONS

## 4a. NORTH ELEVATION

Front view showing 1 storey(s), window/door openings, roof profile and floor levels.

### NORTH ELEVATION - 1 Stories | Style: Modern



EMERSON EIMS | NORTH ELEVATION

Scale 1:50 | 2026-04-26 | Drawing: EL-N01

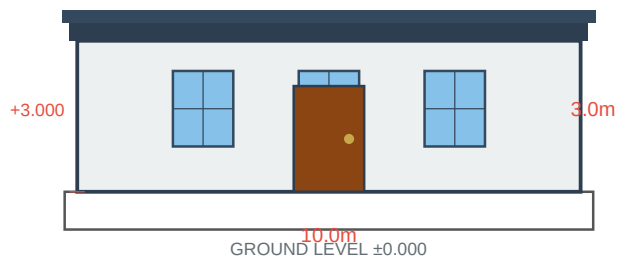
Stories: 1 | Height: 3.0m

Style: Modern | IFC Ready

## 4b. EAST ELEVATION

Side view showing building depth, roof slope and storey heights.

### EAST ELEVATION - 1 Stories | Style: Modern



EMERSON EIMS | EAST ELEVATION

Scale 1:50 | 2026-04-26 | Drawing: EL-E01

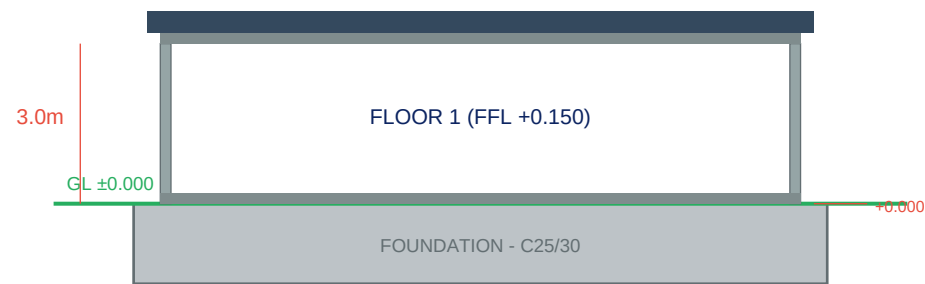
Stories: 1 | Height: 3.0m

Style: Modern | IFC Ready

## 12. BUILDING CROSS SECTION

Vertical cut through building showing floor slabs, storey heights (1 floors), foundation depth, roof structure and structural members.

### CROSS SECTION A-A | 1 Stories



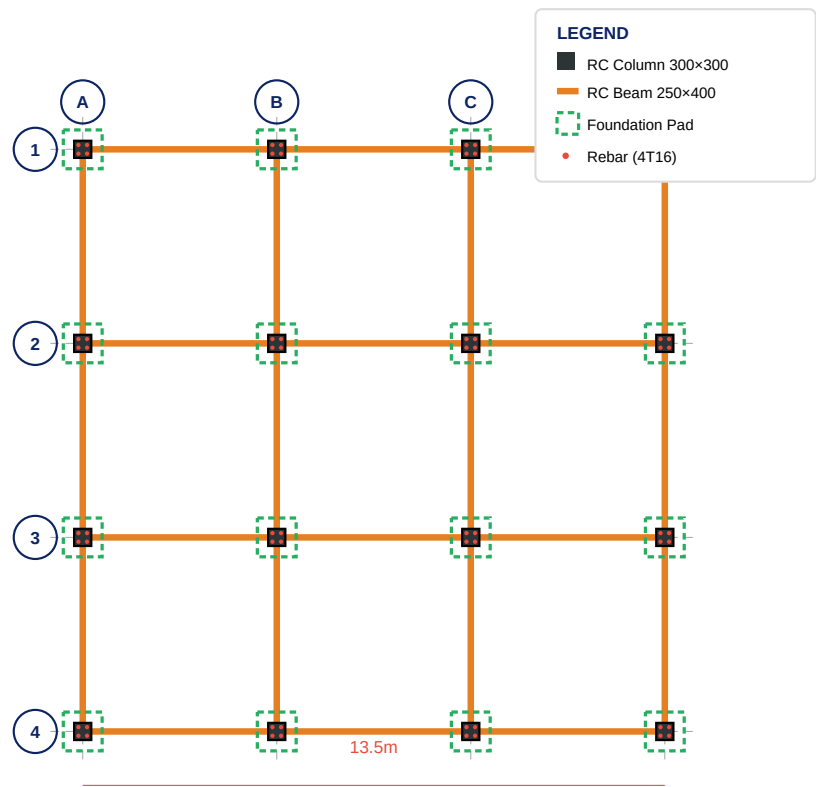
EMERSON EIMS | BUILDING SECTION A-A

2026-04-26 | SEC-001 | Scale 1:50

# 13. STRUCTURAL LAYOUT

Column grid, beam layout and foundation pads. Foundation type: Pad. All dimensions in metres. Concrete grade C25/30, Steel Y16.

## STRUCTURAL LAYOUT - Pad Foundation | 1F | Grid 4.5m×4.5m



EMERSON EIMS | STRUCTURAL LAYOUT

Grid: 4.5m x 4.5m | Columns: 16 | 2026-04-26 | STR-001

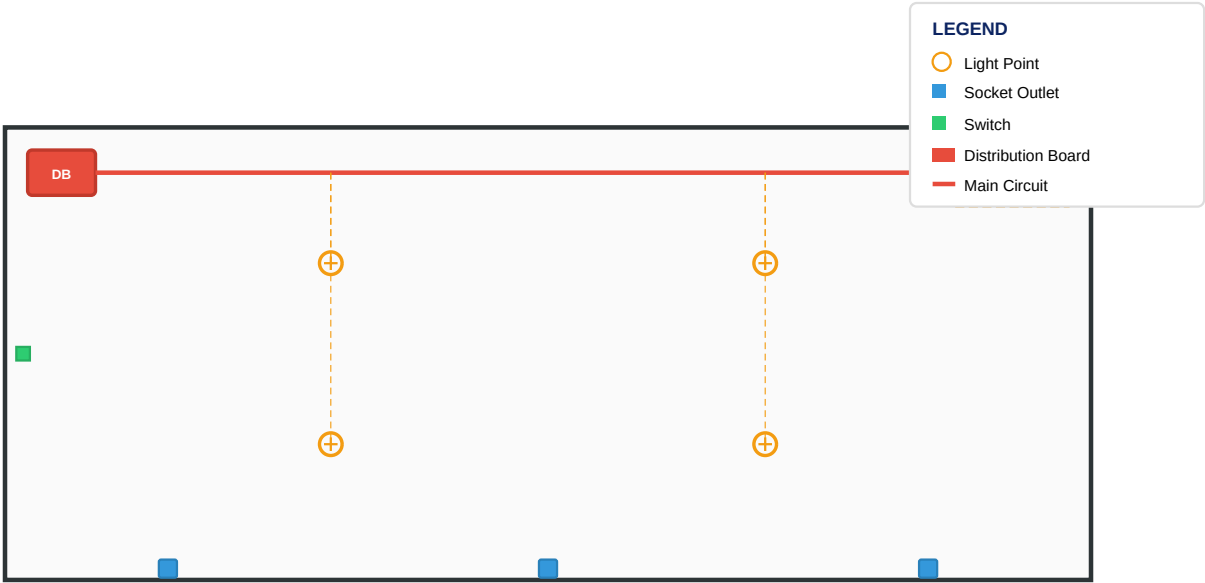
Concrete: C25/30 | Steel: Y16

Foundation: Pad | FOS ≥ 3.0

# 14. MEP LAYOUT — ELECTRICAL

Electrical distribution layout showing DB positions, circuit runs, socket/light points and earthing.

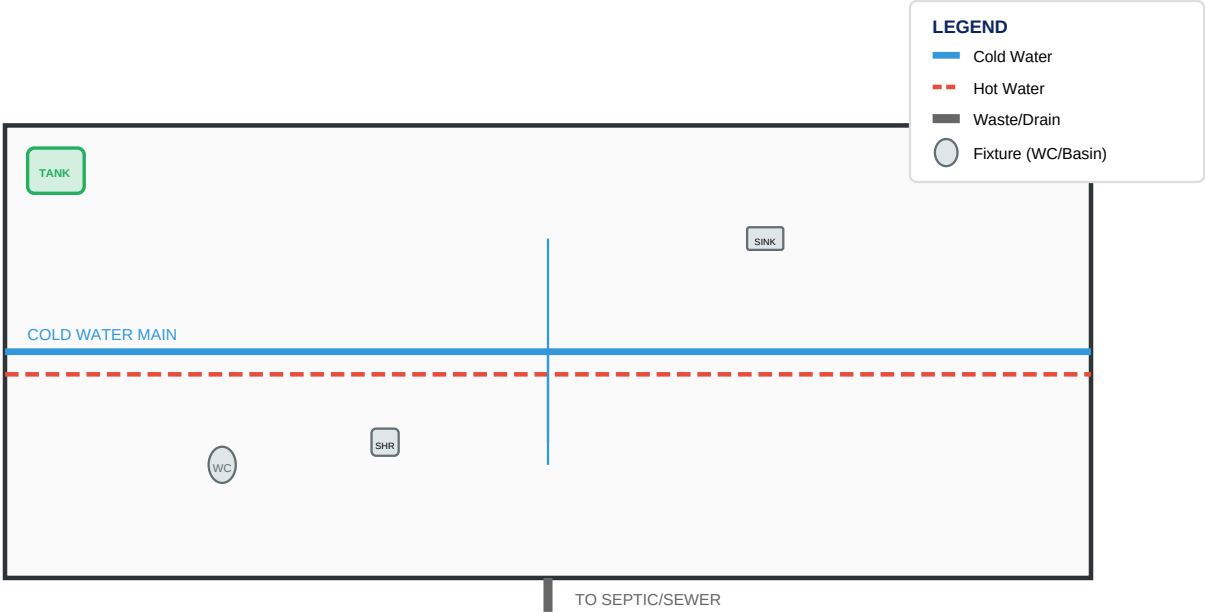
## ELECTRICAL LAYOUT - 100.0m² | 1 Units



# 15. MEP LAYOUT — PLUMBING

Plumbing layout showing supply/drain runs, fixture locations, water heater and main shutoff valve.

## PLUMBING LAYOUT - 100.0m² | 1 Units



## 16. COST SUMMARY

Cost data not yet generated. Run the analysis first.

*This report was generated by EMERSON EIMS Building Suite Pro. All drawings are indicative and must be verified by a licensed engineer before construction.*